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Variable-Fidelity Physics Applied to
Multi-Element Airfoil Optimization**

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FIRST-ORDER MODEL MANAGEMENT WITH VARIABLE-FIDELITY PHYSICS APPLIED TO MULTI-ELEMENT AIRFOIL OPTIMIZATION

N. M. ALEXANDROV*, E. J. NIELSEN†, R. M. LEWIS‡ AND W. K. ANDERSON§

Abstract

First-order approximation and model management is a methodology for a systematic use of variable-fidelity models or approximations in optimization. The intent of model management is to attain convergence to high-fidelity solutions with minimal expense in high-fidelity computations. The savings in terms of computationally intensive evaluations depends on the ability of the available lower-fidelity model or a suite of models to predict the improvement trends for the high-fidelity problem. Variable-fidelity models can be represented by data-fitting approximations, variable-resolution models, variable-convergence models, or variable physical fidelity models. The present work considers the use of variable-fidelity physics models. We demonstrate the performance of model management on an aerodynamic optimization of a multi-element airfoil designed to operate in the transonic regime. Reynolds-averaged Navier-Stokes equations represent the high-fidelity model, while the Euler equations represent the low-fidelity model. An unstructured mesh-based analysis code FUN2D evaluates functions and sensitivity derivatives for both models. Model management for the present demonstration problem yields fivefold savings in terms of high-fidelity evaluations compared to optimization done with high-fidelity computations alone.

Key Words: Aerodynamic optimization, airfoil design, approximation concepts, approximation management, model management, nonlinear programming, surrogate optimization, variable-fidelity

Background

Approximations and low-fidelity models have long been used in engineering design to reduce the cost of optimization (e.g., [1–3]). An overview of approximations in

structural optimization, for instance, can be found in [4]. Accounts of recent efforts in developing methodologies for variable-complexity modeling are relayed in [5, 6].

The present work concerns an approach, the Approximation and Model Management Framework (AMMF) [7–10], designed to enable rapid and early integration of high fidelity nonlinear analyses and experimental results into the multidisciplinary optimization process. This is accomplished by reducing the frequency of performing high-fidelity computations within a single optimization procedure.

Until recently, procedures for the use of variable-fidelity models and approximations in design had relied on heuristics or engineering intuition. In addition, with a few exceptions (e.g., [11], [12]), the analysis of algorithms had focused on convergence to a solution of the approximate or surrogate problem ([13], [14]). The AMMF methodology discussed here and in related papers is distinguished by a systematic approach to alternating the use of variable-fidelity models that results in procedures that are provably globally convergent to critical points or solutions of the high-fidelity problem.

Model management can be, in principle, imposed on any optimization algorithm and used with any models. In [15], we considered AMMF schemes based on three nonlinear programming methods and demonstrated them on a 3D aerodynamic wing optimization problem and a 2D airfoil optimization problem. In both cases, Euler analysis performed on meshes of varying degree of refinement formed a suite of variable-resolution models. Results indicated approximately threefold savings (similar across the three schemes) in terms of high-fidelity function evaluations. The AMMF based on the sequential quadratic programming (SQP) approach was judged to be the most promising for single-discipline problems with a modest

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number of design variables, as well as for certain formulations of the MDO problem.

The study in [15] has served as a proof of concept for AMMF in the case where low-fidelity models are represented by data-fitting approximations (kriging, splines and polynomial response surfaces) or variable-resolution models. The present work considers, arguably, the most challenging combination of the high and low-fidelity models within a single optimization procedure – that of variable-fidelity physics models. The performance of the first-order model management is demonstrated on an aerodynamic optimization of a multi-element airfoil. Variable-fidelity models are represented by an unstructured mesh-based analysis run in viscous and inviscid modes.

In the next section, we describe the AMMF under investigation and discuss the points of interest for the current study. We then present the demonstration problem, followed by a discussion of the numerical experiments and results.

AMMF under investigation

For the present demonstration, the optimal design problem is represented by the bound constrained nonlinear programming problem:

$$\begin{aligned} \min \quad & f(x) \\ \text{s. t.} \quad & l \leq x \leq u, \end{aligned} \quad (1)$$

where x is the vector of design variables, the objective f is a continuously differentiable nonlinear function, and $l \leq x \leq u$ denotes bound constraints on the design variables.

The first-order AMMF used here for solving (1) is based on the trust-region strategy, which is a methodology for the improvement of global behavior of the local model-based optimization algorithms [16]. The following pseudo-code describes the AMMF.

```

Initialize  $x_c, \Delta_c$ 
Do until convergence:
  Select a model  $a_c$  such that
   $a_c(x_c) = f(x_c)$  and  $\nabla a_c(x_c) = \nabla f(x_c)$ 
  Solve approximately for  $s_c = x - x_c$ :
    
$$\begin{aligned} \min_s \quad & a_c(x_c + s) \\ \text{s. t.} \quad & l \leq x \leq u \\ & \|s\|_\infty \leq \Delta_c \end{aligned}$$

  Compute  $\rho_c \equiv \frac{f(x_c) - f(x_c + s_c)}{f(x_c) - a_c(x_c + s_c)}$ 
  Update  $\Delta_c$  and  $x_c$  based on  $\rho_c$ 
End do

```

Details of the updating strategy can be found, for instance, in [7]. Briefly, the point x_c is accepted if the step s_c results in a simple decrease in the objective function, i.e., if

$f(x_c) > f(x_c + s_c)$. Otherwise the step is rejected. The ℓ_∞ trust region (chosen because it interacts naturally with the bound constraints) is decreased if ρ_c is small. Experience suggests that “small” be taken as less than 10^{-5} . If ρ_c is close to one or greater than one (this indicates excellent predictive properties of the model), the trust region is doubled. Otherwise, it is left unchanged.

The conditions

$$a_c(x_c) = f(x_c) \quad (2)$$

$$\nabla a_c(x_c) = \nabla f(x_c) \quad (3)$$

are known as the first-order consistency conditions, which we will discuss presently.

In conventional optimization, a_c is usually a linear or quadratic model of the objective f . AMMF replaces this local, Taylor series approximation by an arbitrary model required to satisfy the consistency conditions (2)–(3). Regardless of the properties of the low-fidelity model, the consistency conditions force it to behave as a first-order Taylor series approximation at points where they are satisfied. Solving the subproblem of minimizing a_c is itself an iterative procedure that now requires the function and derivative information from the low-fidelity model.

First-order AMMF methods can be shown to converge to critical points or solutions of the high-fidelity problem under appropriate standard assumptions of continuity and boundedness of the constituent functions and derivatives (see [9], for instance), given that the consistency conditions (2)–(3) are imposed at each major iterate x_c .

Qualitatively, the reason a first-order AMMF converges to an answer of the high-fidelity problem may be summarized as follows. Although a lower-fidelity model may not capture a particular feature of the physical phenomenon to the same degree of accuracy (or at all) as its higher-fidelity counterpart, a lower-fidelity model may still have satisfactory predictive properties for the purposes of finding a good direction of improvement for the higher-fidelity model. By imposing the consistency conditions, AMMF ensures that at least at the major iterates, the lower-fidelity model provides the same direction of descent as the high-fidelity counterpart. Two questions arise. How easy is it to impose the first-order consistency? How does the method perform in practice?

The answer to the first question is that imposing the conditions (2)–(3) is straightforward using a correction technique due to Chang et al. [17] This technique corrects a low-fidelity version f_{lo} of an arbitrary function so that it agrees to first-order with a given high-fidelity version f_{hi} . This is done by defining the correction factor β as

$$\beta(x) = \frac{f_{hi}(x)}{f_{lo}(x)}.$$

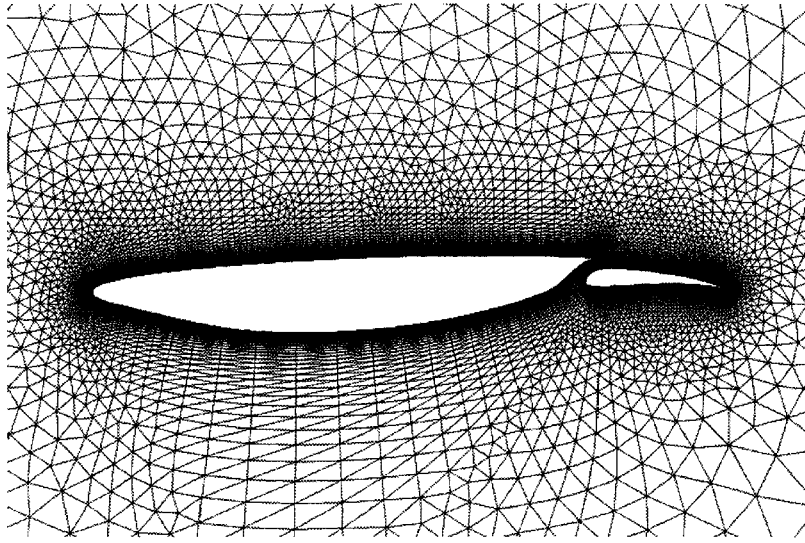


Figure 1: Mesh for the viscous model

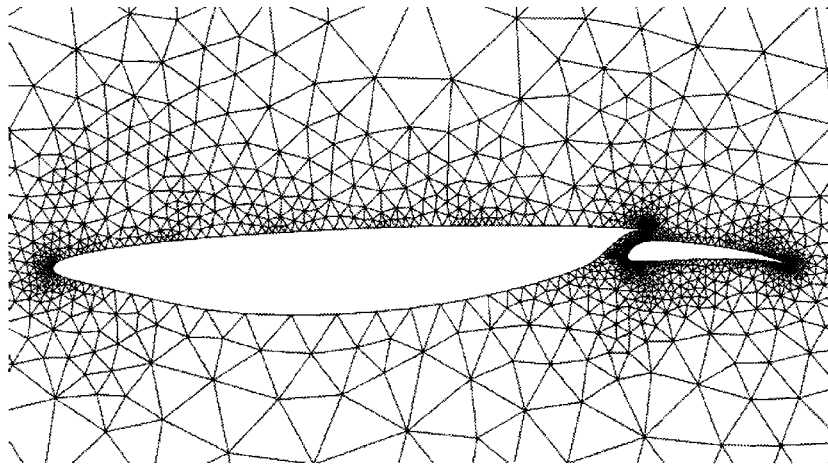


Figure 2: Mesh for the inviscid model

Given the current design variable vector x_c , one builds a first-order model β_c of β about x_c :

$$\beta_c(x) = \beta(x_c) + \nabla \beta(x_c)^T (x - x_c).$$

The local model of β is then used to correct f_{lo} to obtain a better approximation $a(x)$ of f_{hi} :

$$f_{hi}(x) = \beta(x)f_{lo}(x) \approx a(x) \equiv \beta_c(x)f_{lo}(x).$$

The corrected approximation $a(x)$ has the properties that $a(x_c) = f_{hi}(x_c)$ and $\nabla a(x_c) = \nabla f_{hi}(x_c)$. Zero-order or higher-order corrections are easily constructed as well.

Because the β -correction can make any two unrelated functions match to first order, the framework admits a wide range of models. In the worst case of performance, the subproblem will yield a good predictive step s_c for the

high-fidelity model only at the point x_c . Thus, the high-fidelity information may potentially have to be computed at every step to re-calibrate the low-fidelity information. This would lead AMMF to become, at worst, a conventional optimization algorithm. At its best, the AMMF would be able to take many steps with the corrected low-fidelity model before resorting to re-calibration with expensive evaluations. Which scenario actually takes place depends on the problem at hand.

AMMF has shown promise with low-fidelity models represented by data-fitting approximations and variable-resolution models. In an attempt to evaluate the potential worst-case scenario, we are now considering managing variable physical fidelity models. We view this model combination as the potential worst-case scenario, because

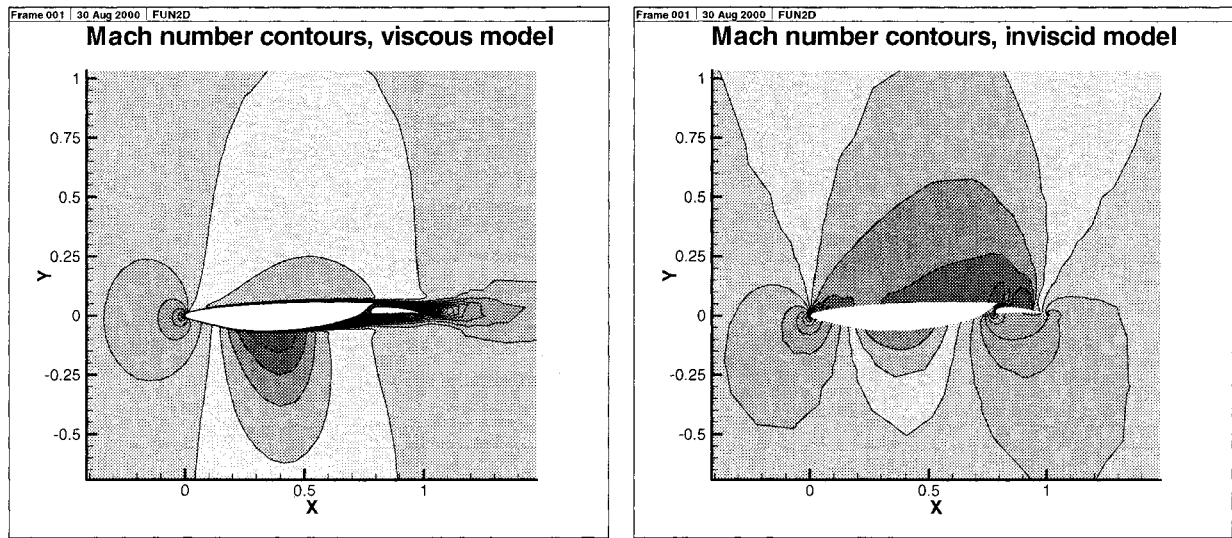


Figure 3: Mach number contours for viscous vs. inviscid model

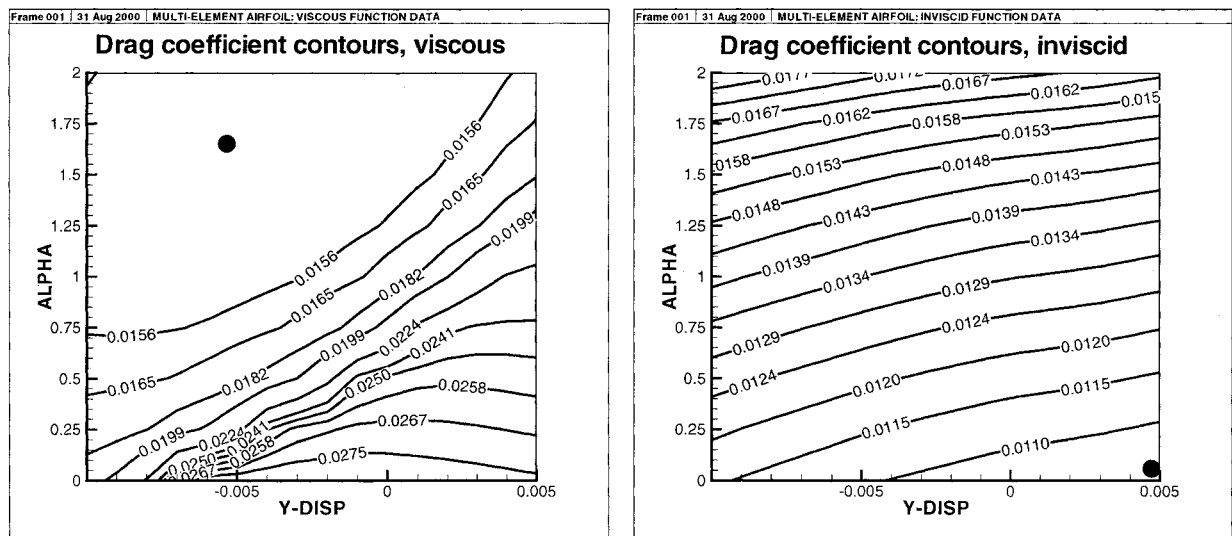


Figure 4: Drag coefficient contours of the viscous and inviscid models

low-fidelity physics models are expected not to capture the behavior of the high-fidelity counterparts accurately, or at all, over some or all regions of interest in the design space.

Demonstration problem

We consider aerodynamic optimization of a two-element airfoil designed to operate in transonic conditions [18]. The inclusion of viscous effects is very important for obtaining physically correct results. Therefore, the high-fidelity model will be the Reynolds-averaged Navier-Stokes equations and the low-fidelity model will

be the Euler equations. The flow solver, FUN2D, used for this study follows the unstructured mesh methodology [19]. Sensitivity derivatives are provided via a hand-coded adjoint approach [20].

The mesh for the viscous model depicted in Fig. 1 consists of 10449 nodes and 20900 triangles. The mesh for the inviscid model, shown in Fig. 2, comprises 1947 nodes and 3896 triangles. The Mach number is $M_\infty = 0.75$, the Reynolds number is $Re = 9 \times 10^6$, the global angle of attack is $\alpha = 1^\circ$.

Fig. 3 depicts the Mach number contours for the viscous and inviscid model, respectively. The boundary and

Test	hi-fi eval	lo-fi eval	total time	factor
PORT with hi-fi surrogates, 2 var	15/15		negligible	
AMMF with surrogates, 2 var	3/3	18/9	negligible	≈ 5
PORT with hi-fi analyses, 2 var	14/13		≈ 12 hrs	
AMMF with direct analyses, 2 var	3/3	19/9	≈ 2.41 hrs	≈ 4.98

Table 1: AMMF performance vs PORT

shear layers are clearly visible in the viscous case. Because of the importance of the viscous effects in this problems, the use of the inviscid equations for the low-fidelity model should present an important test for the present approach.

The objective of this problem is simply to minimize the drag coefficient by adjusting the global angle of attack and the y -displacement of the flap. In this study, we restrict ourselves to two design variables to enable visualization. The baseline case for both models was constructed at $\alpha = 1^\circ$ and zero y -displacement of the flap.

Fig. 4 depicts the level sets of the drag coefficient for the viscous and inviscid models. The problem appears to support the worse-case scenario: not only is the low-fidelity model not a good representation of the high-fidelity model but, in addition, the descent trends in the two models are reversed. The solution for each problem is marked with a circle. Thus the problem provides a good test of the methodology indeed.

The computational expense necessary to calculate functions and derivatives in the viscous case is considerably greater than that for the inviscid model. We conducted our experiments on an SGITM OriginTM 2000 workstation with four MIPS RISC R10000 processors. One low-fidelity analysis took approximately 23 seconds and one low-fidelity sensitivity analysis took between 70 and 100 seconds. In contrast, one high-fidelity analysis took approximately 21 minutes and one high-fidelity sensitivity analysis took between 21 and 42 minutes to compute. The measures were taken in CPU time. Thus, the time per low-fidelity evaluation may be considered negligible compared to that required for a high-fidelity evaluation.

Numerical results

We conducted the following computational experiments. Because our test problem has expensive function evaluations, we first built spline substitutes both for the viscous and the inviscid model. Error analysis indicated that the spline fit was highly satisfactory for both models. It should be emphasized that we did not use these

substitutes in the conventional sense, i.e., they were not used to provide lower-fidelity models. Instead, they simply served to provide low-cost substitutes for both models for the problem components in the testing phase. Of course, such a test would never be conducted in a non-research setting, nor would it be considered for a problem with more than a few variables. In our setting, however, it saved us much time by providing an excellent approximation of the actual functions with respect to descent characteristics at a tiny fraction of computational cost. After we ascertained the correctness of our procedures, tests were conducted directly with the flow and adjoint solver, without recourse to substitutes, because the substitutes were expected to smooth out the problem to a certain degree.

The problems were first solved with single-fidelity models alone by using well-known commercial optimization software[†] PORT [21], in order to obtain a baseline number of function evaluations or iterations to find an optimum. The problems were then solved with AMMF. Identical experiments were conducted with spline substitutes and with the actual flow and adjoint solver.

For each experiment, performance of AMMF was evaluated in terms of the absolute number of calls to the high and low-fidelity function and sensitivity calculations. Because the time for low-fidelity computations was negligible in comparison to the high-fidelity computations, we estimated the savings strictly in terms of high-fidelity evaluations. Table 1 summarizes the number of function (first number) and derivative (second number) computations expended in PORT and in AMMF.

Given the dissimilarity between the high-fidelity and low-fidelity model, we were initially surprised to find that the AMMF performed well: it consistently yielded approximately fivefold savings in terms of high-fidelity computations. The result held both for the spline substitutes and the actual functions. Optimization applied to both cases produced nearly identical iterates.

Following an analysis of the results, we concluded that the savings were not surprising after all. For our combination of models, the β -correction worked extremely well. This is illustrated in Fig. 5. The plot on the left shows the

[†]The use of names of commercial software in this paper is for accurate reporting and does not constitute an official endorsement, either expressed or implied, of such products by the National Aeronautics and Space Administration or Institute for Computer Applications to Science and Engineering.

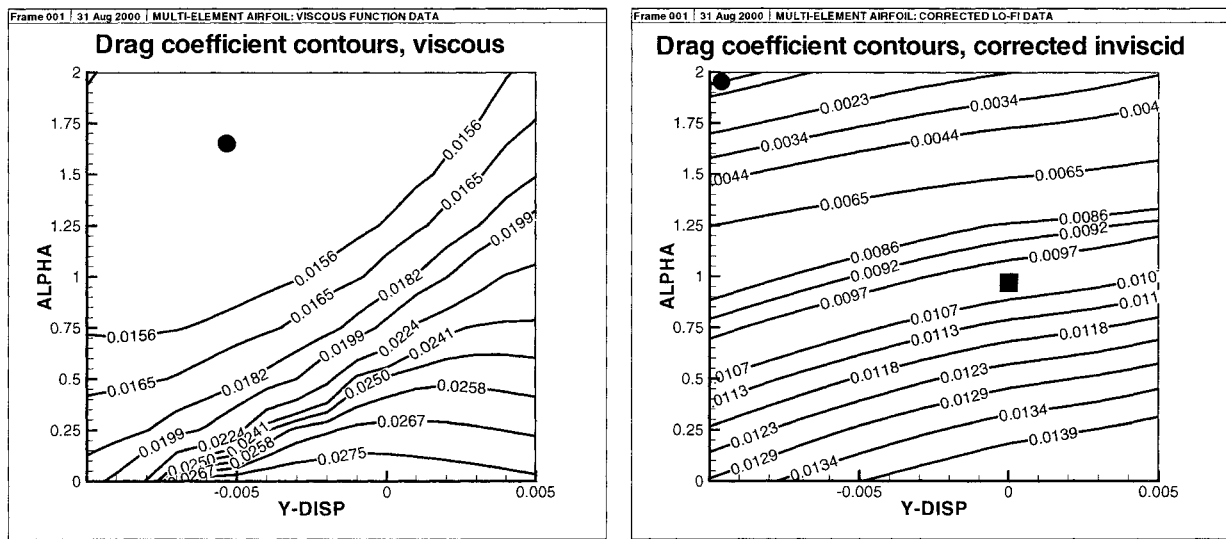


Figure 5: Drag coefficient level sets of the viscous and corrected inviscid models

level sets of the high-fidelity model with the solution. The plot on the right depicts the level sets of the low-fidelity model β -corrected at the initial point. The initial point is marked by a square. We note that the correction is not applied to the entire feasible region during iterations of optimization algorithm. Here, we applied the correction to the entire region to visualize the affect of the correction on the low-fidelity function. The figure clearly shows that the correction, using the function and derivative information at the anchor point (at this iteration - the initial point), reversed the trend of the low-fidelity model, allowing the optimizer to find the next iterate in the left upper corner of the plot, marked by a circle. Similar analysis can be conducted for all iterations. In fact, AMMF located the solution ($\alpha = 1.6305^\circ$, flap y -displacement = -0.0048) of the high-fidelity problem already at the next iteration. The high-fidelity drag coefficient at the initial point was $C_D^{\text{initial}} = 0.0171$, the high-fidelity drag coefficient at the solution was $C_D^{\text{final}} = 0.0148$, a decrease of approximately 13.45%.

Concluding remarks

We believe that the results obtained in this study with AMMF and variable-fidelity physics models are promising. We observed fivefold savings in terms of high-fidelity evaluations compared to conventional optimization. Despite the great dissimilarity between the models, AMMF was able to capture the descent behavior of the high-fidelity model with the assistance of the first-order correction. When the models are greatly dissimilar, first-order information appears indispensable in obtaining reasonable descent directions.

Given the present results, we are cautiously optimistic about several much larger test cases (e.g., 84 variables) that are currently under investigation. Large problems must be tested carefully in AMMF in order to ascertain that its performance is not in some measure an artifact of the problem dimensionality. This does not appear to be the case, because AMMF was previously tested on problems with over ten variables. However, the tests currently conducted with realistic physical models should prove more conclusive, regardless of the outcome.

The performance of AMMF with variable-fidelity physics models raises a number of intriguing questions about the nature of the corrections and an optimal choice of low-fidelity models for a large set of problems. These questions are currently under investigation.

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Findings on the CGI Approach to Enabling Resources on the Internet

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FINDINGS ON THE CGI APPROACH TO ENABLING RESOURCES ON THE INTERNET

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ABSTRACT

In collaborative design, the ability to exchange information among resources (e.g., people, databases, programs, etc.) is essential. The Internet provides a powerful and attractive mechanism to link such resources in a geographically distributed setting. A study summarized in this paper utilizes an accessible web-framework and indicates that a popular resource integration technique using CGI scripts on web servers provides a bottleneck when these resources are automated. This raises concerns about scalability and suggests caution in following this avenue for future engineering system designs. In particular, problems with data persistence and exchange in integrated systems are identified. A lean-server technique, by which an Internet server is embedded directly into engineering system, is shown as one possible solution to the persistence problem. The added benefits of resource coupling and data flow enhancements using content-based markup languages, such as XML, are also explored.

NOMECLATURE

HTML	HyperText Markup Language; a computer language for describing web pages of mixed text, tables, forms and images with embedded references
CGI	Common Gateway Interface; an executable program for running auxiliary processes on web servers
SPACE	System Programming Architecture for Collaborative Engineering; a demonstration framework using lean-server technology.
RBCC	Rocket-Based Combine Cycle; a design framework which combines the most desirable characteristics of air breathing engines and rocket engines into a single, integrated engine
Server	A program that distributes information in response to Internet requests
ASP	Active Server Pages; HTML pages that include server-side scripts
Applet	Java programs which run within HTML pages when viewed by web browsers, which support Java, such as Netscape and Internet Explorer
Servlet	Modules those run inside of a web server to extend its functionality; a server-side applet
Lean-Server	Embodiment of a server directly within a persistent application
XML	eXtensible Markup Language; a W3C proposed recommendation for a file format for distribution of electronic documents on the World Wide Web
DTD	Document Type Definition; used in XML, provides a formal definition of rules for markup

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ENABLING RESOURCES ON THE INTERNET

In a few short years, the Internet has grown from a novelty to the world's library. At the heart of this worldwide information resource is the HyperText Markup Language (HTML). In the context of this paper, HTML provides a language for formatting results for rendering (on a browser screen or printer). It includes standardized markup entities to define visualization, such as fonts, boldface, underlines, headings, tables, forms, frames, as well as a variety of other typesetting properties.

Standalone resources (information, programs, etc.) can be scripted so that their results are output in HTML format and are accessible through a web server. As a result, users and applications in geographically distributed locations can easily harness resources through the Internet. This capability is important because it enables resources to be separately maintained yet be combined to produce larger systems. As an example, a system level engineering analysis tool can be configured from independently derived sub-system analysis coded by separate entities.

There are a number of methods for integrating resources into the Internet through servers. These are described here. The authors have studied the first, the CGI technique, in detail in an attempt to determine a scalable integration solution. The findings will be summarized later in the paper.

Common Gateway Interface

Common Gateway Interface (CGI) is a popular technique for integrating standalone resources on the Internet. The process by which resources are integrated through CGI is depicted in Figure 1. CGI behaves in the following manner:

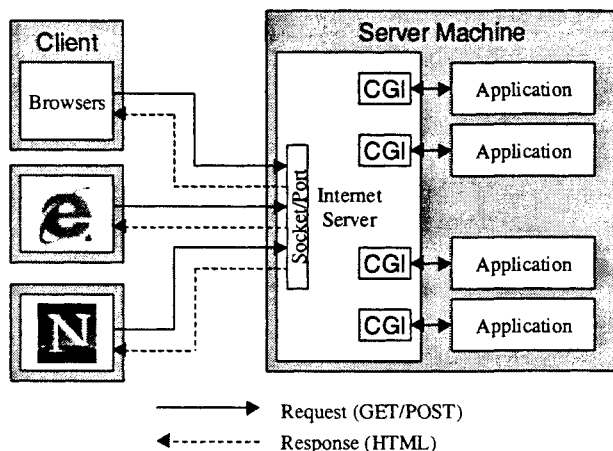


Figure 1. Using CGI to Couple Browsers and Analysis Codes

1. receives client-side requests from the server;
2. executes server-side analysis codes, typically scripted; and
3. returns results, typically in HTML format, to client browsers.

Standard server configurations invoke the CGI process in conjunction with an open browser request for script execution. Information is passed through the CGI as standard input or coalesced into a single shell variable.

Active Server Pages

Active Server Pages (ASP) is similar to CGI in that scripts are deployed on the server. They can be treated as dynamic HTML pages. In the case of ASP, the scripts are written in Microsoft Visual Basic. Since all of the scripts in an Active Server Page are processed on the server before they are sent to a browser, an Active Server Page is compatible with all HTML-compliant browsers. When a browser receives a page generated from an Active Server Page, the browser receives standard HTML. All the work has been performed on the server.

Reading and writing text files or accessing a database in a client-side script, such as applet, raises security concerns. In contrast, you can easily access the file system or a database within an Active Server Page. Since the scripts in an Active Server Page can be kept private, accessing the file system or a database does not introduce the same type of security considerations from authenticated Internet requests.

Servlets

Servlets are to servers what applets are to browsers. Unlike applets, however, servlets have no graphical user interface. Servlets are modules that extend web server functionality. For example, a servlet might be responsible for taking data in an HTML data-entry form and applying engineering programs used perform specific analysis.

Servlets can be embedded in many different servers because the servlet API, which you use to write servlets, assumes nothing about the server's environment or protocol. Servlets have become most widely used within HTTP servers; many web servers support the Servlet API.

Servlets are an effective replacement for CGI scripts. They provide a way to generate dynamic documents that is both easier to write and faster to run. Servlets also address the problem of doing server-side

programming with platform-specific application programming interfaces: they are developed with the Java Servlet API, a standard Java extension.

FRAMEWORKS ON THE INTERNET

The use of CGI as an integration method is a popular method to integrate analysis tools into a web server because it is straightforward to implement for engineering resources. One of the reasons is that the scripting technique is convenient for the integration of file-based standalone analysis tools. Another reason is that the implementer requires little knowledge of newer programming paradigms.

Rocket-Based Combine-Cycle Framework

Two framework systems described in references [1] and [2] both utilize CGI scripts to integrate analysis tools into the environment. The latter is a Rocket-Based Combine-Cycle (RBCC) vehicle design integration framework developed at the Georgia Institute of Technology. As highlighted in Figure 2, the framework consists of four analysis tools:

- Propulsion Analysis,
- Trajectory Analysis,
- Weights and Sizing Analysis, and
- Economics and Cost Estimation.

Some of the tools are executed on UNIX machines and others on the Macintosh platform. The system uses files for data sharing among the applications (persistence). This system illustrates the benefits to be obtained by using CGI-based integration. The key benefit is the integration of legacy programs on disparate platforms achieved through the creation of custom interfaces, which can be used by distributed team members.

The flat file-based, data exchange for the four analysis tools in the RBCC framework is shown in Figure 3. Data are obtained either by soliciting input from users using forms from a web page or through the execution of one of the analysis tools. In the latter cases, the tools are executed in a pre-defined sequence, and data should be exchanged internally between the programs by means of CGI-interaction. This will be discussed in more detail in the following section.

The RBCC tool as described here will be used as a baseline for the study of CGI-based frameworks because it is a good illustration of this approach and readily available via the Internet. The framework is studied in detail and the methods presented in this paper

will show how the investment in this class of framework can be leveraged to build larger systems.

Aircraft Weight and Performance Analysis

Another example of a web-enabled analysis program is described here that also employs CGI technique and executes aircraft weight and performance analyses. This program is developed by the lead author and is implemented in a multi-threaded environment. The following three analysis processes are executed in parallel:

- *Weight Ratio Analysis.* Determines Fuel weight ratio available, fuel weight ratio required for mission, useful load fraction, mission fuel weight balance.
- *Achievable Climb Gradient Analysis.* Calculates take-off achievable climb gradient aborted-landing achievable climb gradient.
- *Operation Field Length Calculation.* Finds take-off field length, and landing field length

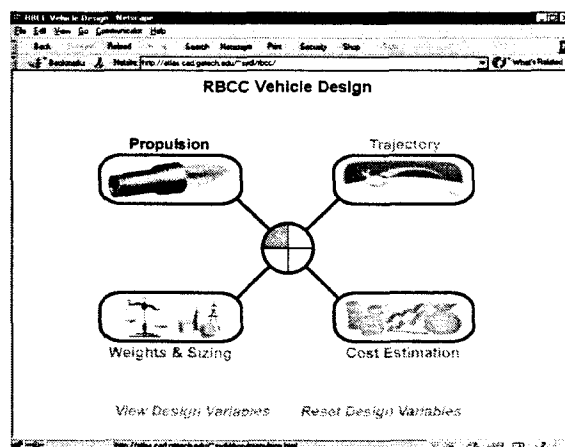


Figure 2. RBCC Vehicle Design Integration Framework

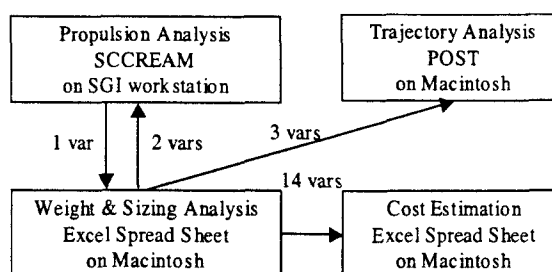


Figure 3. Data Coupling in RBCC Framework

Aircraft Weight & Performance Analysis

Data Input

No. of Engines:	1	Thrust:	40000
Aerodynamic Input Filename:	aero_in.dat	Wing Area:	1700
Take-Off Weight:	300000	Cruise velocity:	778
Fuel Load:	40000	Flight Range:	2900
Fixed Equipment weight:	1000	Engine Specific Fuel Consumption:	0.000194

Analysis

Calculation Results

Take-off Field Length:	18302.286 ft	Landing Field Length:	10811.765 ft
Take-off Achievable Climb Gradient:	0.061	Aborted landing Achievable Climb Gradient:	0.080
Fuel Weight Ratio Available:	0.339	Fuel Weight Ratio Required for Mission:	0.164
Lifted Load Fraction:	0.297	Mission Fuel Weight Balance:	1.293

Figure 4. Aircraft Weight and Performance Analysis

When a user enters input data in the appropriate fields and clicks the "Analysis" button as shown in Figure 4, client data is transferred to the web server. A server-side CGI script then extracts data and inserts them into program input files. Finally, the programs are executed in parallel using multi-threading. After analysis is finished, CGI returns results to client-side web browser via an HTML file.

ON TO COLLABORATION

Collaboration is one of the drivers of the next generation of Internet applications. It brings a grand vision of designers working at geographically distributed locations, making better design decisions in a reduced time frame, lowering product cost, and achieving revolutionary concepts never thought possible before [3]. To turn this into reality, integrated design frameworks operating in a heterogeneous environment is needed. At the same time, collaborative activities are built on the exchange of information among different resources (e.g., people, databases, programs, etc.); this means that exchanging information on the Internet is fundamental to collaboration.

CGI Resources in Collaboration

Considerable effort is being expended to enable individual software programs on the Internet using the CGI-technique. As described earlier, the reason for this is the simplicity of the method yielding widespread usage. Penalties arise when these CGI-enabled applications are used in information intensive collaboration projects. This occurs because the

intermixing of visual formatting markups blocks the data extraction of analysis results for subsequent tool use. Collaboration capable frameworks that invoke several CGI scripts need to extract variable values from HTML results, which are the solutions returned by several legacy applications, in order to pass the values to other programs in the process. The extraction requires additional "wrappers" to parse needed information from results formatted for visual display; this can be an erroneous and inefficient process.

Study Using the Tools of the RBCC Framework

The four analysis tools that are implemented in the RBCC design framework described earlier were used in a baseline implementation by the authors. This is done to illustrate where the pitfalls occur in information intensive design systems. A schematic of the resulting framework is depicted in Figure 5 and is designed to:

- execute the CGI-scripts in-place as to leverage the original investment,
- exchange the data as shown in Figure 3 to mimic a collaboration problem, and
- execute the tools in the proper sequence so that the variables are calculated as needed instead of being obtained from users in a manual process.

The resulting framework implementation was found to be highly problem dependent and difficult to program. The problems were encountered when the CGI resources were used "as-is" to demonstrate a scalable platform as required for collaboration. Recall that individual CGI responses are HTML-formatted results

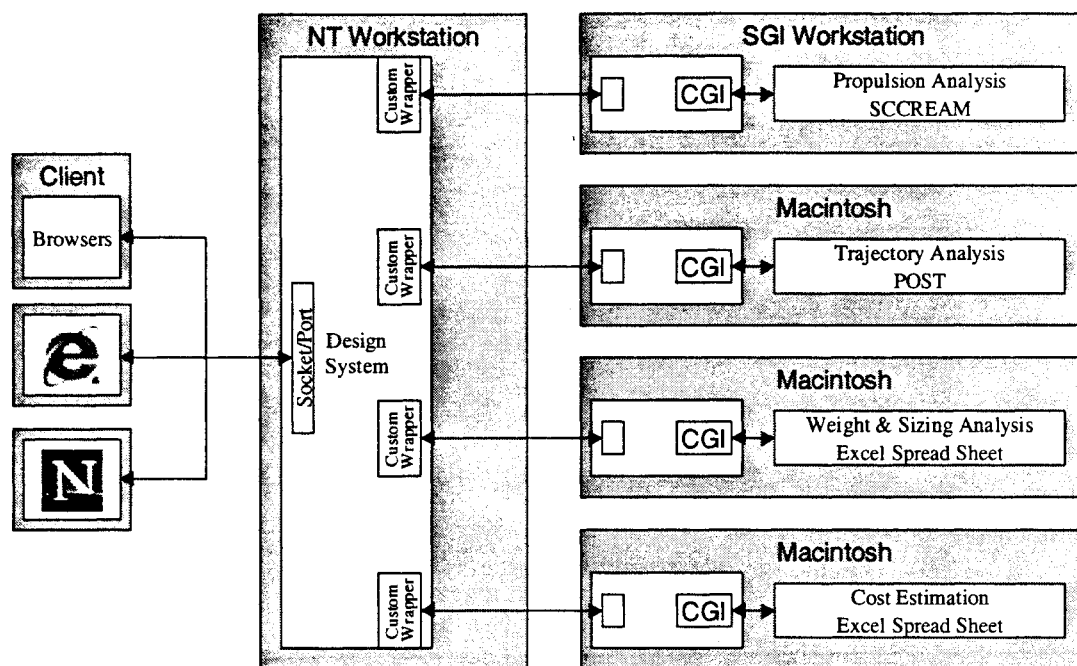


Figure 5. CGI Integration with Wrappers

for depicting solutions to the user in a web browser. Hence the data extraction process is a considerable barrier for automation and limits re-usability.

The second problem with this framework is less obvious. It is that each resource is bound to a CGI connection that cannot be maintained across sessions. If a Web page is stopped from loading in its entirety or the user leaves that page, the executing CGI process will stop to execution. Each CGI-enabled application is executed in its own process space to maintain secure. This presents a lack of inter-session persistence that is needed in collaboration and cannot be achieved using CGI.

STREAMLINING COLLABORATION

Having identified a significant bottleneck in the deployment of CGI-based frameworks, attention is turned to making these frameworks more efficient. It is important to note that the CGI implementations described here work well for solving individual problems. It is when they are deployed in collaborative frameworks that the problems occur. Most of these problems can be overcome if the tools were implemented from a framework perspective at the onset. To bridge this gap, a simplified execution scheme is first presented for using legacy resources. Second, a data-centric markup language is argued to facilitate data exchange.

Persistence

Persistence of applications is a problem in CGI-based frameworks because the scripts are bound to the connection channel. It should be pointed out that a majority of the file-based CGI wrappers studied by the authors to date have static filenames and, subsequently, do not permit more than one connection to operate uniquely.

Several alternatives exist that permit server data persistence. These include the following.

- Connect application to server. For example, use servlets, or extend server.
- Build application as a server. Embed Internet request/response capabilities into an application to enable the resource to behave as a server.

As a practical implementation, a new lean-server technique, which represents the latter approach, is described here as an application-level method for embodying persistence.

Implementing a Lean-Server Enabled System to Facilitate Persistence

The use of a lean-server is an approach that embodies an Internet server directly within a persistent design system or application. The lean-server is described in detail in Ref 4, including instructions for its implementation. In short, the functional elements of

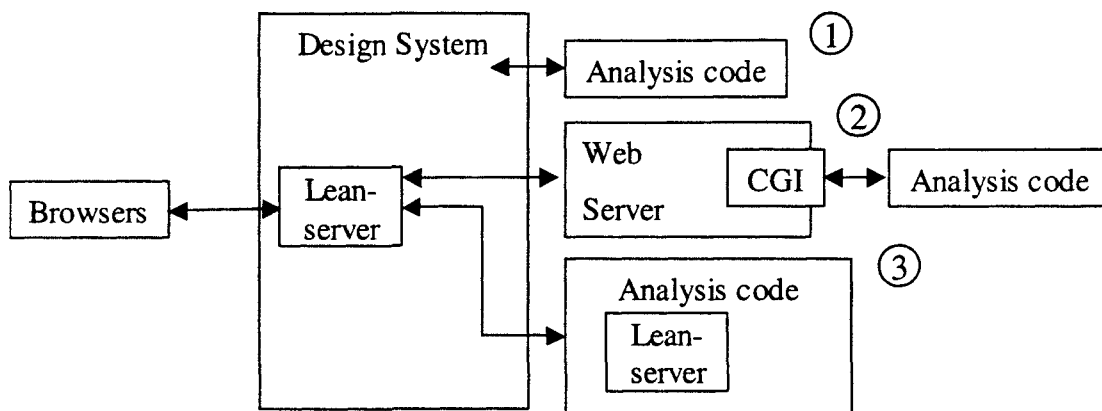


Figure 6. Lean-server Approach

popular web servers are coalesced into a single interface. This interface can in-turn be embedded into other applications. Thus, design frameworks, scripts, and individual programs can each be made into a web server. This enables each application to be accessed directly across the Internet!

The study here will focus on a centralized design system being critical to the implementation of a collaborative environment. This system is responsible for user interfaces, resource allocation, and data archival. It provides the persistence crucial for collaboration. Furthermore, the design system will have a lean-server imbedded in it to enable the system to be directly accessible via the Internet.

Individual applications can be interfaced to the design system in a number of ways as shown in Figure 6.

1. Applications can be invoked using traditional methods such as exec and system calls found in most procedural languages.
2. Applications can be accessed using a GET or POST HTTP request over a socket connection to a web server. This would be the case for applications that are already implemented as CGI scripts as described earlier. The lean-server includes interfaces to streamline this process.
3. Applications can themselves be lean-server enabled and each application interfaces directly through its lean-server. This enables applications to be persistent and behave with properties similar to a design system if managed correctly. These

applications are analogous to peers in a peer-peer framework as often seen in high-end simulation. In this case, control is delegated to the individual applications.

The net result of the use of a lean-server enabled application is that connectivity to the web is largely simplified. And each program, which is standalone before, can be accessed as an independent unit via Internet without server involved.

Simplified Data Persistence in the Lean-Server

As a final note, the lean-server approach not only simplifies the framework structure, but also benefits the data transfer process itself. The lean-server maintains an internal hash table (variable-value combinations) for maintaining data flow when interfacing with other resources using one of the methods as shown in Figure 6. This table would normally be coupled with a back-end archival system for data management tasks. The hash facility can be leveraged in combining CGI scripts as done in the RBCC demonstration.

The hash table can be used to store intermediate values when coupling applications. The table can be combined with intermediary scripts to facilitate the use of CGI-enabled applications that the design system may control. This process is shown in Figure 7. The lean-server contains built-in functions for aiding the end-user in integrating applications with the hash table. In the RBCC demonstration, the hash table is used to manage data entered by the end-user on Web page Forms as well as data created by the coupling of the analysis tools.

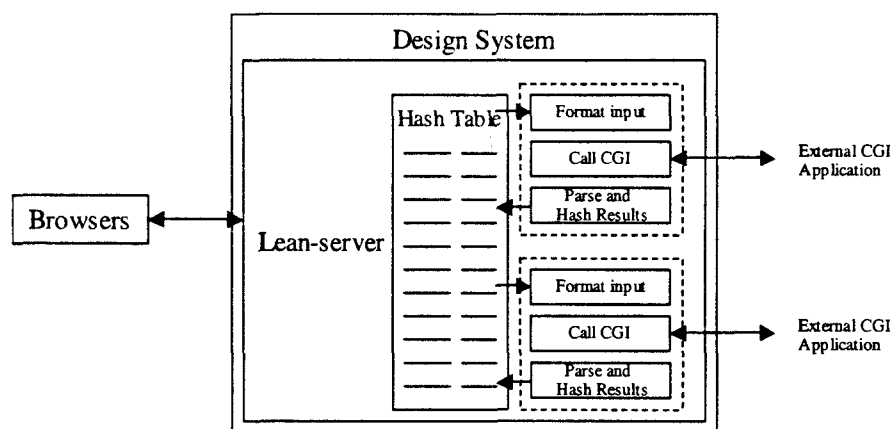


Figure 7. Hash Table use in Lean-Server Applications

Revealing Data Structure using XML

Re-visiting the desire for facilitating collaboration on the Internet. The goal is to achieve high-end data communication using the resources. A potential barrier exists in the responses from CGI-enabled applications typically coded in HTML. As has been shown here, these responses need to distill to their elementary variables for use in larger systems in which the processes are inter-connected. *Would it not be easier to return the results in a data-centric format?* This is the foundation of the eXtensible Markup Language (XML).

XML is a specification for how to organize structured data into document [6]. It is simple, flexible and powerful. The following simple example shows the difference between HTML (Figure 8) and XML (Figure 9). In HTML, the <p> and <h1> tags are used to indicate paragraph separation for browser display. In contrast, XML tags come in pairs with each marking the beginning and ending of an element, a piece of information. Elements may include atomic information or include other elements. For example, the element <TelephoneBill> is a complex element, called the root element, and it surrounds all document information. The element <To> is a simple element that captures a piece of atomic data, the name of the person for the bill. Finally, an XML document with properly nested tags is called well-formed.

XML supports the notion of a Document Type Definition (DTD) [7,8], which specifies a particular arrangement of elements within an XML document. For example, the DTD for a company database might specify that each company database consists of a company element containing one or more employee elements, and that each employee element contains a name, an age, and a phone number in exactly that order. An XML document is considered valid if it conforms to a particular DTD, and an XML-based application can

validate a document to ensure that it has an appropriate structure before using it.

While HTML describes the way data should look, XML provides a language for data structure. This capability is desired for the category of system that is the focus of this paper: interoperable analysis tools in a design system. The use of XML provides a method for identifying content that is more robust than parsing because analysis tools identify the information as it is created.

```

<!--HTML Script -->
<h1> Telephone Bill </h1>
<p> From: BellSouth Corporation
<p> To: Tom Jones
<p> Date: Jan. 1, 2000
<p> Amount: $20.00
<p> Tax: 5%
<p> Total Due: $21.00

```

Figure 8. Sample Address Entry in HTML

```

<!--XML script -->
<TelephoneBill>
  <From>BellSouth Corporation</From>
  <To>Tom Jones</To>
  <Date>9/15/2000</Date>
  <Amount Currency='Dollars'>20.00</Amount>
  <TaxRate>5</TaxRate>
  <TotalDue Currency='Dollars'>21.00</TotalDue>
</TelephoneBill>

```

Figure 9. Sample Address Entry in XML

A Final Note on Symmetric Processes and Modern Standards-Based Design Systems

Following the pitfalls of CGI systems and some of the possible solutions, it is possible to suggest that collaboration be focused on symmetric (request and response structures use similar data encapsulation), data-centric applications. That is to shift in implementing the CGI-enabled applications as described in this paper and shown in Figure 10-1 to a combined XML and lean-server architecture as shown in Figure 10-2. This architecture design has the following properties:

- data is easily interchanged and understood among applications because it is XML;
- applications are persistent across sessions because they are lean-server enabled; and
- control is simplified because both request and responses are in the same XML format, and server with its related components is not needed.

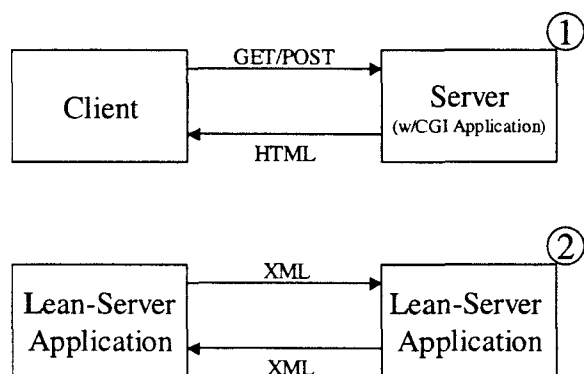


Figure 10. Symmetric Data-Centric Applications

There are a number of standards communities embracing findings in this paper but from a business perspective. These include the ebXML [9] (focused on business process meta-models), RosettaNet [10] (focused on business processes), and Microsoft BizTalk™ [11] (focused on trading partners). This is important because investment in linking engineering resources to the Internet needs to be leveraged to be interoperable in these environments as well.

SUMMARY

The use of CGI as an integration method is a popular method to integrate analysis tools into a web server because it is straightforward to implement for engineering resources. The reasons for this are the simplicity of the method yielding widespread usage. Penalties arise when these CGI-enabled applications are used in information intensive collaboration projects. This occurs because the intermixing of visual formatting markups blocks the data extraction of analysis results for subsequent tool use. The problem is compounded as design moves to new collaborative environments that invoke several CGI scripts must extract solutions from HTML results, which are returned by several legacy applications.

Analysis tools in an existing RBCC framework were re-used in a design framework for study of the CGI re-use problem. The resulting framework implementation was found to be highly problem dependent and difficult to program. The problems were encountered when the CGI resources were used "as-is" to demonstrate a scalable platform as required for collaboration. Hence the data extraction process is a considerable barrier for automation and limits re-usability. The second problem with this framework is that each resource is bound to a CGI connection that cannot be maintained across sessions.

Two concepts for overcoming some of the barriers of CGI tool reuse were introduced in this paper. The first, the lean-server technique, is an approach that embodies an Internet server directly within a persistent design system or application. Thus, design frameworks, scripts, and individual programs can each be made into a web-server. The second, the use of XML, is an approach whereby data elements are identified. Thus data reuse is facilitated. When these elements are combined, the authors noted similarity from standards-based frameworks developed from a business modeling perspective.

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